

CHAPTER III.6, WEEK 1

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6.1. Let $(X, \mathcal{O}_X)$ be a ringed space, and let $\mathcal{F}', \mathcal{F}'' \in \mathfrak{Mod}(X)$. An extension of $\mathcal{F}''$ by $\mathcal{F}'$ is a short exact sequence

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$$

in $\mathfrak{Mod}(X)$. Two extensions are *isomorphic* if there is an isomorphism of the short exact sequences, inducing the identity maps on $\mathcal{F}'$ and $\mathcal{F}''$. Given an extension as above, consider the long exact sequence arising from $\text{Hom}(\mathcal{F}'', \cdot)$, in particular the map

$$\delta : \text{Hom}(\mathcal{F}'', \mathcal{F}'') \rightarrow \text{Ext}^1(\mathcal{F}'', \mathcal{F}'),$$

and let $\xi \in \text{Ext}^1(\mathcal{F}'', \mathcal{F}')$ be $\delta(1_{\mathcal{F}''})$. Show that this process gives a one-to-one correspondence between isomorphism classes of extensions of $\mathcal{F}''$ by $\mathcal{F}'$, and elements of the group $\text{Ext}^1(\mathcal{F}'', \mathcal{F}')$. For more details, see, e.g., Hilton and Stammach, Ch. III.

Proof. (1) We define a map from $\text{Ext}^1(\mathcal{F}'', \mathcal{F}')$ to extensions $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{E} \rightarrow \mathcal{F}'' \rightarrow 0$ as follows. Fix an injective resolution $0 \rightarrow \mathcal{F}' \xrightarrow{d^{-1}} \mathcal{I}^\bullet$ of $\mathcal{F}'$ with maps $d^i : \mathcal{I}^i \rightarrow \mathcal{I}^{i+1}$, so that

$$\begin{aligned} \text{Ext}^1(\mathcal{F}'', \mathcal{F}') &\cong h^1(\text{Hom}(\mathcal{F}'', \mathcal{I}^\bullet)) \\ &\cong \frac{\ker d^1}{\text{im } d^0}. \end{aligned}$$

Thus, we can regard an element $[\xi] \in \text{Ext}^1(\mathcal{F}'', \mathcal{F}')$ as the class of a map $\xi : \mathcal{F}'' \rightarrow \mathcal{I}^1$ with $d^1 \xi = 0$. Fix such an element $[\xi]$ with representative ξ , and take the pullout $\mathcal{P} := \mathcal{I}^0 \times_{\mathcal{I}^1} \mathcal{F}''$ in the following diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}' & \xrightarrow{d^{-1}} & \mathcal{I}^0 & \xrightarrow{d^0} & \mathcal{I}^1 & \xrightarrow{d^1} & \mathcal{I}^2 \\ & & & & \uparrow \gamma & & \uparrow \xi & & \\ & & & & \mathcal{P} & \xrightarrow{\psi} & \mathcal{F}'' & & \end{array}$$

Then we have $d^0 \phi = 0$, so the universal property of $\mathcal{P}$ supplies a unique map $\phi : \mathcal{F}' \rightarrow \mathcal{P}$ such that the following diagram commutes:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}' & \xrightarrow{d^{-1}} & \mathcal{I}^0 & \xrightarrow{d^0} & \mathcal{I}^1 & \xrightarrow{d^1} & \mathcal{I}^2 \\ & & \nearrow d^{-1} & & \uparrow \gamma & & \uparrow \xi & & \\ & & \mathcal{F}' & \xrightarrow{\exists! \phi} & \mathcal{P} & \xrightarrow{\psi} & \mathcal{F}'' & & \\ & & & \searrow & & & & & \\ & & & & 0 & & & & \end{array}$$

We will now show that $0 \rightarrow \mathcal{F}' \xrightarrow{\phi} \mathcal{P} \xrightarrow{\psi} \mathcal{F}'' \rightarrow 0$ is a short exact sequence.

ϕ is monic since $d^{-1} = \gamma\phi$ is monic, and $\psi\phi = 0$ by construction. To show exactness at the middle, it suffices to show that $\text{im } \phi$ satisfies the universal property of $\ker \psi$. Let $\alpha : \mathcal{A} \rightarrow \mathcal{P}$ be a morphism such that $\psi\alpha = 0$. It suffices to show that α factors through $\text{im } \phi$. Recall that ϕ, d^{-1} are monic and $\mathcal{F}' \rightarrow \mathcal{I}^0 \rightarrow \mathcal{I}^1$ is exact, so

$$\text{im } \phi \cong \text{coim } \phi \cong \mathcal{F}' \cong \text{im } d^{-1} \cong \ker d^0.$$

Then $d^0\gamma\alpha = \xi\psi\alpha = 0$, so the universal property of kernels supplies a unique map $\tilde{\alpha} : \mathcal{A} \rightarrow \mathcal{F}'$ such that $\gamma\alpha = d^{-1}\tilde{\alpha}$. Further, $\gamma\phi\tilde{\alpha} = \gamma\alpha$ and $\psi\phi\tilde{\alpha} = 0 = \psi\alpha$ imply that $\alpha = \phi\tilde{\alpha}$ via universal property of pullbacks.

$$\begin{array}{ccccc} & & \mathcal{I}^0 & \xrightarrow{d^0} & \mathcal{I}^1 & \xrightarrow{d^1} & \mathcal{I}^2 \\ & \nearrow d^{-1} & \uparrow \gamma & \lrcorner & \uparrow \xi & & \\ \mathcal{F}' & \xrightarrow{\phi} & \mathcal{P} & \xrightarrow{\psi} & \mathcal{F}'' & & \\ & \nwarrow \exists! \tilde{\alpha} & \uparrow \alpha & \nearrow 0 & & & \\ & & \mathcal{A} & & & & \end{array}$$

Hence, the above diagram commutes, and $\tilde{\alpha}$ is indeed unique since ϕ is monic.

Finally, we show that ψ is epic. $d^1\xi = 0$ allows us to factor ξ through $\ker d^1 = \text{im } d^0$, giving us the following diagram, where the rightmost square is a pullback.

$$\begin{array}{ccccc} \mathcal{I}^0 & \xrightarrow{\tilde{d}^0} & \text{im } d^1 & \xrightarrow{i} & \mathcal{I}^1 \\ \gamma \uparrow & & \uparrow \tilde{\xi} & \lrcorner & \uparrow \xi \\ \mathcal{P} & \xrightarrow{\psi} & \mathcal{F}'' & \xrightarrow{\text{id}} & \mathcal{F}'' \end{array}$$

It follows from the following elementary lemma on pullbacks that the lefthand square is a pullback as well.

Lemma 1. *Suppose that the rightmost square is a pullback. Then the leftmost square is a pullback if and only if the entire rectangle is a pullback.*

$$\begin{array}{ccccc} \bullet & \longrightarrow & \bullet & \longrightarrow & \bullet \\ \uparrow & & \uparrow & \lrcorner & \uparrow \\ \bullet & \longrightarrow & \bullet & \longrightarrow & \bullet \end{array}$$

ψ epic will follow from the following lemma.

Lemma 2. *Pullbacks in an abelian category preserve epimorphisms.¹*

Proof. Consider the following pullback square.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ a \uparrow & \lrcorner & \uparrow b \\ A' & \xrightarrow{f'} & B' \end{array}$$

¹I went on a day long wild goose chase trying to prove this with a more standard diagram chase using that all epimorphisms in an abelian category are cokernels. I originally thought it was a general fact that coequalizers are preserved by pullbacks, but that didn't last long, thankfully.

We are working in an abelian category, so that the following sequence is left exact.

$$0 \longrightarrow A' \xrightarrow{\begin{pmatrix} a \\ f' \end{pmatrix}} A \oplus B' \xrightarrow{(f \ -b)} B \longrightarrow 0$$

f epic implies $(f \ -b)$ is epic, so that the sequence is right exact as well. Hence, our original square was bicartesian, and f' must be epic since pushforwards preserve epimorphisms in an arbitrary category. $\square$

(2) Now, we must show the extension $0 \rightarrow \mathcal{F}' \xrightarrow{\phi} \mathcal{P} \xrightarrow{\psi} \mathcal{F}'' \rightarrow 0$ depends only on the cohomology class of ξ . First, suppose $\xi = d^0\zeta$ for some $\zeta \in \text{Hom}(\mathcal{F}'', \mathcal{I}^0)$, and consider the following pullback square.

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}' & \xrightarrow{d^{-1}} & \mathcal{I}^0 & \xrightarrow{d^0} & \mathcal{I}^1 \xrightarrow{d^1} \mathcal{I}^2 \\ & & \nearrow \gamma' & & \uparrow \gamma & \nwarrow \zeta & \uparrow \xi \\ & & & & \mathcal{P} & \xrightarrow{\psi} & \mathcal{F}'' \\ & & \searrow \psi' & & & & \end{array}$$

Let $\xi, \xi' \in \text{Hom}(\mathcal{F}'', \mathcal{I}^1)$ be cohomologous maps, and take $\zeta \in \text{Hom}(\mathcal{F}'', \mathcal{I}^0)$ such that $\xi = \xi' + d^0\zeta$. Then it consider the following diagram.

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}' & \xrightarrow{d^{-1}} & \mathcal{I}^0 & \xrightarrow{d^0} & \mathcal{I}^1 \xrightarrow{d^1} \mathcal{I}^2 \\ & & \nearrow \gamma' & & \uparrow \gamma & \nwarrow \zeta & \uparrow \xi \\ & & & & \mathcal{P} & \xrightarrow{\psi} & \mathcal{F}'' \\ & & \searrow \psi' & & & & \end{array}$$

We have $\xi\psi = \xi'\psi + d^0\zeta\psi =$ $\square$

6.2. Let $X = \mathbb{P}_k^1$, with k an infinite field.

- Show that there does not exist a projective object $\mathcal{P} \in \mathbf{Mod}(X)$, together with a surjective map $\mathcal{P} \rightarrow \mathcal{O}_X \rightarrow 0$. [Hint: Consider surjections of the form $\mathcal{O}_V \rightarrow k(x) \rightarrow 0$, where $x \in X$ is a closed point, V is an open neighborhood of x , and $\mathcal{O}_V = j_!(\mathcal{O}_X|_V)$, where $j : V \rightarrow X$ is the inclusion.]
- Show that there does not exist a projective object $\mathcal{P}$ in either $\mathbf{Qcoh}(X)$ or $\mathbf{Coh}(X)$ together with a surjection $\mathcal{P} \rightarrow \mathcal{O}_X \rightarrow 0$ [Hint: Consider surjections of the form $\mathcal{L} \rightarrow \mathcal{L} \otimes k(x) \rightarrow 0$, where $x \in X$ is a closed point, and $\mathcal{L}$ is an invertible sheaf on X .]

Proof. (a) Suppose not, and take such a $\mathcal{P} \in \mathbf{Mod}(X)$ and surjective map $\phi : \mathcal{P} \twoheadrightarrow \mathcal{O}_X$. Take a closed point $x \in X$, and let $U, V \ni x$ be neighborhoods of x such that $U \not\subset V$ and $V \not\subset U$. Then we have a surjection $j_!(\mathcal{O}_X|_V) =: \mathcal{O}_V \twoheadrightarrow k(x) := i_*(k(x))$, and hence the

following diagram via the projective lifting property.

$$\begin{array}{ccccc}
 \mathcal{O}_V & \longrightarrow & k(x) & \longrightarrow & 0 \\
 & \nearrow & \uparrow & & \\
 & & \mathcal{O}_X & & \\
 & & \uparrow \phi & & \\
 & & \mathcal{P} & &
 \end{array}$$

Now, take $s \in \Gamma(U, \mathcal{P})$. We have $\Gamma(U, \mathcal{O}_V) = 0$ since $V \not\subset U$, so $\phi(s) \mapsto 0$ under $\mathcal{O}_X \rightarrow k(x)$. But then $\text{im } \phi_x \subset m_x \subsetneq \mathcal{O}_{X,x}$, contradicting surjectivity of ϕ .

(b) Suppose not, and take such a $\mathcal{P} \in \mathfrak{Mod}(X)$ and surjective map $\mathcal{P} \rightarrow \mathcal{O}_X \rightarrow 0$. Fix $x \in X$ and consider the surjection $\mathcal{O}_X \rightarrow k(x) \rightarrow 0$. (Ex. 5.10) gives us $n > 0$ such that both $\Gamma(X, \mathcal{O}_X(n)) \rightarrow k(x) \rightarrow 0$ and $\Gamma(X, \mathcal{P}(n)) \rightarrow \Gamma(X, \mathcal{O}_X(n)) \rightarrow 0$ are surjective. Tensoring $\mathcal{O}_X \rightarrow k(x) \rightarrow 0$ with $\mathcal{O}_X(-n-1)$ gives us a surjection

$$\mathcal{O}_X(-n-1) \rightarrow k(x) \otimes \mathcal{O}_X(-n-1) \cong k(x) \rightarrow 0$$

Once again, the projective criterion gives us a map $\mathcal{P} \rightarrow \mathcal{O}_X(-n-1)$ such that the following diagram commutes.

$$\begin{array}{ccccc}
 \mathcal{O}_X(-n-1) & \longrightarrow & k(x) & \longrightarrow & 0 \\
 & \nearrow & \uparrow & & \\
 & & \mathcal{O}_X & & \\
 & & \uparrow & & \\
 & & \mathcal{P} & &
 \end{array}$$

We can tensor this diagram with $\mathcal{O}_X(n)$.

$$\begin{array}{ccccc}
 \mathcal{O}_X(-1) & \longrightarrow & k(x) & \longrightarrow & 0 \\
 & \nearrow & \uparrow & & \\
 & & \mathcal{O}_X(n) & & \\
 & & \uparrow & & \\
 & & \mathcal{P}(n) & &
 \end{array}$$

Then the following diagram commutes.

$$\begin{array}{ccc}
 0 = \Gamma(X, \mathcal{O}_X(-1)) & \longrightarrow & k(x) \neq 0 \\
 \nwarrow & & \uparrow \\
 & & \Gamma(X, \mathcal{O}_X(n)) \\
 \nearrow & & \uparrow \\
 & & \Gamma(X, \mathcal{P}(n))
 \end{array}$$

However, this implies $0 = \Gamma(X, \mathcal{O}_X(-1)) \rightarrow k(x) \neq 0$ is a surjection, which is a contradiction. $\square$

6.3. Let X be a noetherian, and let $\mathcal{F}, \mathcal{G} \in \mathfrak{Mod}(X)$.

- (a) If $\mathcal{F}, \mathcal{G}$ are both coherent, then $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$ is coherent, for all $i \geq 0$.
- (b) If $\mathcal{F}$ is coherent and $\mathcal{G}$ is quasi-coherent, then $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$ is quasi-coherent, for all $i \geq 0$.

Proof. (a) We begin by showing the result for $i = 0$. Take an affine open $\text{Spec } A \cong U \subset X$ and finitely generated A -modules M, N such that $\mathcal{F}|_U \cong \widetilde{M}$ and $\mathcal{G}|_U \cong \widetilde{N}$. Then we have

$$\text{Hom}_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{G}|_U) \cong \text{Hom}_A(M, N).$$

We claim that $\text{Hom}_A(M, N)$ is a finitely generated A -module. Take presentations $A^l \rightarrow A^m \rightarrow M \rightarrow 0$ and $A^l \rightarrow A^n \rightarrow N \rightarrow 0$, so that the following diagram commutes.

$$\begin{array}{ccc} A^{m \times n} \cong \text{Hom}(A^m, A^n) & \longrightarrow & \text{Hom}(A^m, N) \\ & & \uparrow \\ & & \text{Hom}(M, N) \end{array}$$

Then $\text{Hom}(M, N)$ is a submodule of a finitely generated module, and thus finitely generated since A is noetherian.

In particular, we can take a distinguished affine $D(f) \subset \text{Spec } A$ for $f \in A$, and see via (6.2) that

$$\text{Hom}_{\mathcal{O}_{D(f)}}(\mathcal{F}|_{D(f)}, \mathcal{G}|_{D(f)}) \cong \text{Hom}_{A_f}(M_f, N_f).$$

We claim that

$$\text{Hom}_{A_f}(M_f, N_f) \cong \text{Hom}_A(M, N)_f,$$

whence it would follow from the above that

$$\mathcal{H}om_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{G}|_U) \cong \widetilde{\text{Hom}_A(M, N)}.$$

Fix presentations $A^l \rightarrow A^m \rightarrow M \rightarrow 0$ and $A^l \rightarrow A^n \rightarrow N \rightarrow 0$ as before. Clearly the result holds when M, N are both free modules, as

$$\text{Hom}_A(A^m, A^l)_f \cong A_f^{m \times l} \cong \text{Hom}_{A_f}(A_f^m, A_f^l).$$

Localization is exact, so the following sequence is exact,

$$\text{Hom}_{A_f}(A_f^m, A_f^l) \longrightarrow \text{Hom}_{A_f}(A_f^m, A_f^n) \longrightarrow \text{Hom}_A(A^m, N)_f \longrightarrow 0$$

whence

$$\text{Hom}_A(A^m, N)_f \cong \frac{\text{Hom}_{A_f}(A_f^m, A_f^n)}{\text{Hom}_{A_f}(A_f^m, A_f^l)} \cong \text{Hom}_{A_f}(A_f^m, N_f).$$

That is, the result holds when M is free. Finally, the same trick gives us exactness of the following diagram,

$$0 \longrightarrow \text{Hom}_A(M, N)_f \longrightarrow \text{Hom}_{A_f}(A_f^n, N_f) \longrightarrow \text{Hom}_{A_f}(A_f^l, N_f)$$

so that

$$\text{Hom}_A(M, N)_f \cong \ker(\text{Hom}_{A_f}(A_f^n, N_f) \rightarrow \text{Hom}_{A_f}(A_f^l, N_f)) \cong \text{Hom}_{A_f}(M_f, N_f).$$

We conclude that $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ is indeed coherent.

Finally, we show that $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$ is coherent for all i . X is noetherian and $\mathcal{F}$ is coherent, so we can take an exact sequence $0 \rightarrow \mathcal{K} \rightarrow \mathcal{E} \rightarrow \mathcal{F} \rightarrow 0$ with $\mathcal{E}$ locally free and $\mathcal{K}$ coherent. Take the corresponding long exact sequence.

$$\begin{aligned} 0 &\longrightarrow \mathcal{H}om(\mathcal{F}, \mathcal{G}) \longrightarrow \mathcal{H}om(\mathcal{E}, \mathcal{G}) \longrightarrow \mathcal{H}om(\mathcal{K}, \mathcal{G}) \longrightarrow \cdots \\ \cdots &\longrightarrow \mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) \longrightarrow \mathcal{E}xt^i(\mathcal{E}, \mathcal{G}) \longrightarrow \mathcal{E}xt^i(\mathcal{K}, \mathcal{G}) \longrightarrow \cdots \end{aligned}$$

(6.7) and (6.3) give us that

$$\mathcal{E}xt^i(\mathcal{E}, \mathcal{G}) \cong \mathcal{E}xt^i(\mathcal{O}_X, \mathcal{E}^\vee \otimes \mathcal{G}) = 0$$

for all $i > 0$, so the connecting maps are in fact isomorphisms

$$\mathcal{E}xt^i(\mathcal{K}, \mathcal{G}) \cong \mathcal{E}xt^{i+1}(\mathcal{F}, \mathcal{G}).$$

Recall that $\mathcal{K}$ is coherent, and thus $\mathcal{E}xt^0(\mathcal{K}, \mathcal{G}) \cong \mathcal{H}om(\mathcal{K}, \mathcal{G})$ is coherent. The result holds by induction on i .

(b) $\mathcal{G}$ coherent was only used to show that $\mathcal{H}om_A(M, N)$ is finitely generated, so we are done. $\square$

6.5. Let X be a noetherian scheme, and assume that $\mathbf{Coh}(X)$ has enough locally free (Ex. 6.4). Then for any coherent sheaf $\mathcal{F}$ we define the *homological dimension* of $\mathcal{F}$, denoted $\text{hd}(\mathcal{F})$, to be the least length of a locally free resolution of $\mathcal{F}$ (or $+\infty$ if there is no finite one). Show:

- (a) $\mathcal{F}$ is locally free $\iff \mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) = 0$ for all $\mathcal{G} \in \mathbf{Mod}(X)$;
- (b) $\text{hd}(\mathcal{F}) \leq n \iff \mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) = 0$ for all $i > n$ and all $\mathcal{G} \in \mathbf{Mod}(X)$;
- (c) $\text{hd}(\mathcal{F}) = \sup_x \text{pd}_{\mathcal{O}_x} \mathcal{F}_x$.

Proof. f

$\square$

6.8.